Tax Productivity and Development: A Quantitative Macroeconomic Analysis

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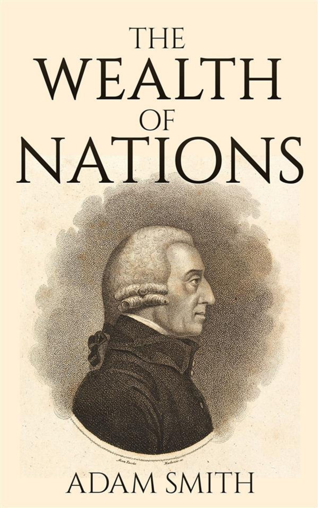
The Importance of Tax Collection

- Core task of government; prerequisite for any public spending.

- Central to **'state capacity'** theory of development,

  Besley and Persson (2011), Besley, Ilzetzki, and Persson (2013), Mayshar, Moav, and Pascali (2022), Tilly (1990), others.

Of Taxes in *The Wealth of Nations*



"*Every tax ought to be so contrived as both to take out and to keep out of the pockets of the people as little as possible over and above what it brings into the public treasury of the state...* ***the cost of collecting it may be very considerable, and in the worst cases may amount to more than it brings in.***"

Q: Why Was Egypt Great?

Egyptian Pyramids ≈ 3000 BCE

New Guinea sweet potatoes ≈ 5000 BCE

A: The Nilometer. Mayshar, Moav, Pascali (2021)'s Appropriability

Open Questions:

- Tax collection capacity is not easy to measure,
  - Typically proxied by Taxes/GDP, but quite indirect measure
  - Theories of state capacity $\Rightarrow$ tax capacity increases Taxes and GDP.
- Little consensus on *quantitative importance* of tax collection capacity,
  - **Q1**: How much long-run GDP per capita increase if tax collection capacity in LICs raised?
    - **Q2**: Can firms meaningfully benefit from such tax collection capacity increases?

Overview of This Paper

- **Data**: new data on tax collection across 100 countries
  - Tax collections (T) / collection eXpenditures (X)
  - Document robust & strong positive correlation with GDP per capita in:
    - (1) cross-section, (2) time series, and (3) across local states

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- **Framework**: quantitative dynamic G.E. model to interpret data + simulate counterfactuals
  1. **Tax administration** in workhorse Ramsey model
  2. **Public capital** complimentary to private capital à la Barro (1990), baxter-King (1993)
  3. Reallocation to and from **modern/traditional sector**

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- We define notion of **tax productivity**

- $\downarrow$ tax productivity $\Rightarrow \downarrow$ public capital $\Rightarrow \downarrow$ firm productivity $\Rightarrow \downarrow$ GDP per capita

Key Results

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  - **Order of magnitude larger** than simple "Hulten" logic, which suggests $+0.3$ %

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- **A1**: 3x African LIC's **tax productivity would boost LR GDP per capita** by $\approx +3.7$ %
  - **Order of magnitude larger** than simple "Hulten" logic, which suggests $+0.3$ %
- **A2**: **Sizeable part of gains** (80%) explained by **firms' reoptimization**

Empirical Findings: Tax Collections / Expenditures

Measuring Tax Collections/ Expenditure

- Unit of Observation: tax administration

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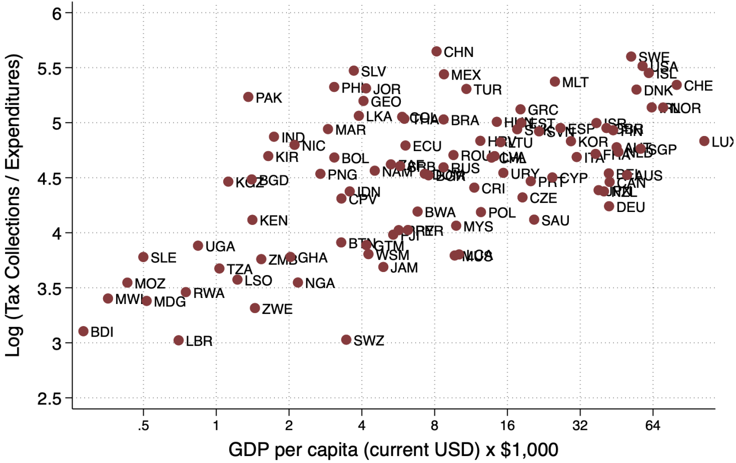
Measuring Tax Collection / Expenditure

- Data Source
  - National tax authority documents
  - IMF Revenue Administration Fiscal Information Tool

- 100 countries, 625 country-year observations
- $T =$ Tax Collections including PIT, CT, VAT, sales, and customs taxes
- $X =$ Official tax collection authority budget
- Tax collection over collection expenditure, T/X
  1. Cross-country comparisons and regressions
  2. Historical Panel Regressions
  3. Sub-national states regressions

1. Tax Collection / Expenditure Increases with GDP per capita

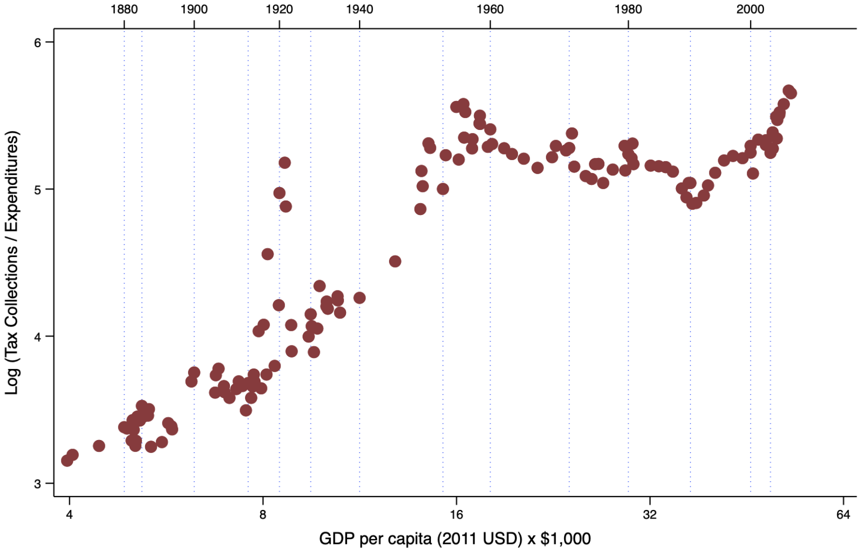


1. Descriptive Cross-Country Regressions: T/X and GDP Per Capita

Outcome	(1) Log T/X	(2) Log T/X	(3) Log T/X	(4) Log T/X	(5) Log T/X	(6) Log T/X	(7) Log T/X	(8) Log T/X
Log GDPpc	0.248*** (0.0389)	0.654*** (0.142)	0.225*** (0.061)	0.221*** (0.0586)	0.249*** (0.0395)	0.260*** (0.0398)	0.218*** (0.0484)	0.241*** (0.0399)
N	100	35	65	99	97	90	73	98
R^2	0.293	0.390	0.174	0.321	0.296	0.326	0.222	0.276
Modif. to column (1)		Only low, lower-middle inc. countries	Only upper-middle, high inc. countries	Cont. for PIT, VAT, CIT, tariff	Excl. if no VAT or sales tax	Excl. if no customs	Excl. if include soc. sec.	Excl. if resp for pop registry

Statutory Tax Rates

2. Tax Collection / Expenditure in the US: 1867 to 2017



2. Descriptive Panel Regressions over 16 Countries

	(1) Log (T/X)	(2) Log (T/X)	(3) Log (T/X)	(4) Log (T/X)
		Exclude if not responsible for customs duties	Exclude if responsible for social security contributions	Exclude if changes in responsibilities
Log Real GDP pc	0.637*** (0.175)	0.636*** (0.197)	0.684*** (0.163)	0.644*** (0.193)
Observations	625	546	519	578
R-squared	0.563	0.541	0.574	0.541
Country FE	Yes	Yes	Yes	Yes

Australia (1972-2017); Bangladesh (1964-2015); Botswana (1971-2005); Canada (1924-2008); Denmark (1959-2006); India (1946-2016); Kenya (1962-1995); Malawi (1964-1996); Pakistan (1949-2005); South Korea (1987-2016); Sierra Leone (1964-1996); South Africa (1960-2007); Thailand (1959-2017); United Kingdom (1993-2017); United States (1877-2016); Uruguay (1980-2017).

3. Sub-National States Regressions

Outcome	(1) Log T/X	(2) Log T/X
Log GDP percap	0.567***	0.278***
	(0.143)	(0.0606)
Sample	US states (1993-2017)	Indian states (1950-2017)
Number of states	50	28
State FE	Yes	Yes
N	850	964
R^2	0.719	0.777

(1) US states (1993-2016), (2) Indian states (1950-2017)

Takeaways

- **T/X positively correlated with GDP per capita**
  1. across countries, robust to tax admin purview & rates
  2. over time
  3. across subnational states

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- Suggestive evidence, but need **structural model** to:
  - Formalize notion of tax productivity
  - Measure tax productivity in cross section of countries (future work)
  - Simulate effects of counterfactually raising tax productivity

Model of Tax Productivity and Development

Overview of Model Environment

- Three novel model features to quantify workhorse Ramsey model

  Judd (1985), Chamley (1986), Straub and Werning (2020), Auclert, Cai, Rognlie, and Straub, (2024),

  1. **Tax administration** in workhorse Ramsey model
     - One input: tax collectors L_C
     - Output: tax rate on modern sector wages, $\tau = G(A^C, L_C)$
     - Tax productivity A^C is exogenous, captures "tax collection capacity"
  2. **Public capital** complimentary to private capital

     Musgrave (1959), Barro (1990), Baxter-King (1993), Weinzeirl (2018)
  3. Reallocation to and from **modern/traditional sector**

Technology: Modern and Traditional Sectors

- **Modern Sector:** production function with public capital k_G**, taxed wages**:

$$Y_M = F(A^M, k_G, k, L_M),$$

hire private (wage $w_{M,t}$) and rent price capital (rate r_t)

$$\max_{k_t,\, L_{M,t}} F(A^M, k_{G,t}, k_t, L_{M,t}) - w_{M,t} L_{M,t} - r_t k_t.$$

- **Traditional Sector:** linear production function, **untaxed**:

$$Y_T = A^T L_T,$$

wage $w_T = A^T$.

Public Sector and Government

- **Tax Collection:** Taxes T_t are collected on modern sectors wages at rate $G(A^C, L_{C,t}) \leq 1$,

$$T_t = G(A^C, L_{C,t}) w_{M,t} L_{M,t}.$$

- **Government budget:** can borrow to finance budget.

$$w_{M,t} L_{C,t} + R_t b_t + k^G_{t+1} \leq T_t + b_{t+1} + (1 - \delta^G) k^G_t.$$

  Public capital evolves as:

$$k_{G,t+1} = (1 - \delta^G)\, k_{G,t} + T_t.$$

  Government cannot borrow to finance public capital.

Households

- Modern and traditional sector workers form a **representative household**

$$\max_{\{c_t, k_{t+1}, b_{t+1}\}} \sum_{t=0}^{\infty} \beta^t \, u(c_t, L_{M,t}, L_{T,t})$$

- **Budget constraint**

$$c_t + k_{t+1} + b_{t+1} \leq (1 + r_t - \delta) k_t + (1 - \underbrace{G(A^C, L_{C,t})}_{\text{tax rate}}) w_{M,t} L_{M,t} + w_T L_{T,t} + R_t \, b_t.$$

Initial private capital $k_0 = \bar{k}_0$.

Equilibrium

A competitive equilibrium is a set of allocations

$$\{c_t, L_{T,t}, L_{M,t}, L_{C,t}, k_t, k_{G,t}, b_t\}$$

and prices

$$\{w_{T,t}, w_{M,t}, r_t, R_t\},$$

such that:

1. **Firm Optimality:**
   - Modern firms take prices and $k_{G,t}$ as given, and maximize profit.
   - Traditional firms hire labor at wage $w_T = A^T$.
2. **Household Optimality:** The household maximizes utility for given prices and L_C.
3. **Budget constraints:** Government BC and public capital accumulation equations hold.
4. **Market Clearing:** The goods market clears:

$$c_t + k_{t+1} + \underbrace{w_{M,t} L_{C,t}}_{\text{collection personnel wages}} + k_{G,t+1} \leq (1 + r_t - \delta) k_t + w_{M,t} L_{M,t} + A^T L_{T,t} + (1 - \delta^G) k_{G,t}.$$

Planner's Problem

- Government chooses the sequence of personnel $\pi = \{L_{C,t}\}_{t=0,\ldots,\infty}$ to maximize welfare:

$$\max_{\pi} \sum_{t=0}^{\infty} \beta^t \, u(c_t(\pi), L_{M,t}(\pi), L_{T,t}(\pi)),$$

subject to:

- the aggregate resource constraint,
- an **implementability condition** for consistency with household optimization,
- the government's budget constraint.

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Lemma

The implementability condition on the planner's allocation choice is:

$$\sum_{t=0}^{\infty} \beta^t \Big[u_c(t) c_t - u_{L_M}(t) L_{M,t} - u_{L_T}(t) L_{T,t} \Big] = u_c(0) \, R_0 \, k_0 .$$

Optimal Tax Collection

At an interior steady state, the optimal tax collection choice satisfies:

$$G_{L_C}(A^C, L_C) = (1-(1-\delta_G)\beta)\,\frac{1}{L_M}\,\frac{F_k+1-\delta}{(F_G-\delta_G)-(F_k-\delta)}.$$

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Economic Interpretation:

- A high F_k (marginal return on private capital) favors private investment over public hiring.
- A high F_G (marginal return on public capital) enhances the attractiveness of public sector spending.
- A higher tax collection productivity A^C improves the effectiveness of tax collectors.
- A larger modern sector (L_M) expands the taxable base.

All endogenous objects.

Quantitative Analysis

Quantitative Analysis

- Goal:
  - Infer tax productivity A^C across countries
  - Simulate long-run aggregate effects of raising tax productivity
  - Simulate transition dynamics
- Today:
  - Proof of concept calibration
  - Long-term effects of doubling tax productivity in a country like Ghana

Functional Forms

- We set the following functional forms:

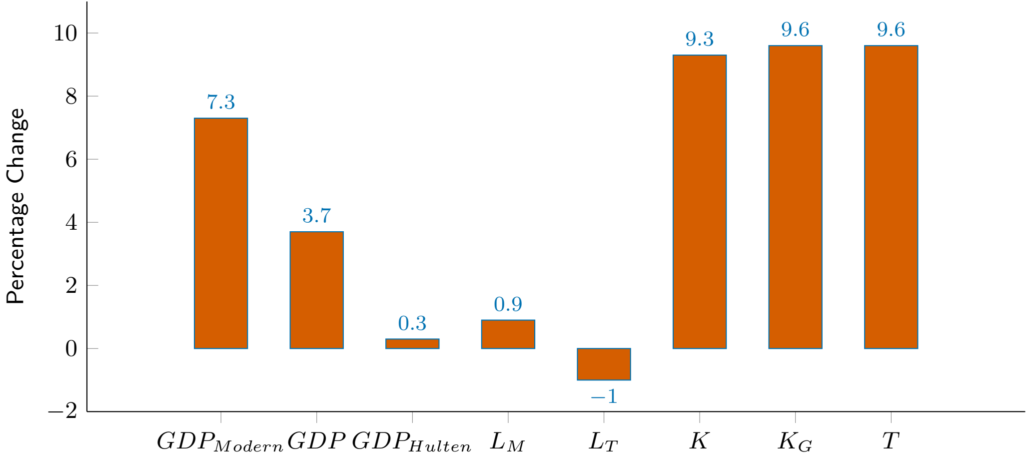
$$
\begin{aligned}
\text{[diminishing marginal products]} \quad & G(A^C, L_C) = A^C L_C^{\eta}, \\
\text{[Barro-Baxter-King complementary]} \quad & F_M(k^G, k, L_M) = A_M (k^G)^{\zeta} k^{\mu} L_M^{1-\mu}, \\
\text{[linear traditional sector]} \quad & F_T(L_T) = A_T L_T, \\
\text{[CRRA]} \quad & u(c) = \frac{c^{1-\gamma}}{1-\gamma}, \\
\text{[isoelastic disulity of labor]} \quad & u(L) = \frac{(L)^{1/\epsilon}}{1+1/\epsilon}.
\end{aligned}
$$

- We introduce private capital wedges $\mu_{k,t}$ in the household's capital decision to capture:
  - realistic distortions in emerging markets
  - potential distortionary effects of public capital.

Experiment: Raise Productivity (A^C) of Ghana

See also Jawara, Knebelmann, Levine, Ndiaye, and Pouliquen (2025) [In the field. The Gambia]

Impact of 3X Tax Productivity



- Domar weight of public sector: $w_M L_C/\text{GDP} = 0.3\%$
- Hulten's theorem $\Rightarrow$ predicted ΔGDP $\approx +0.3\%$
- An order of magnitude larger GDP growth $+3.7\%$

Growth Accounting: Where do the Gains Come From?

$$\Delta \log Y = \underbrace{s_M}_{\textbf{share of modern}} \times \left(\zeta \underbrace{\Delta \log k^G}_{\textbf{public capital growth}} + \mu \underbrace{\Delta \log k}_{\textbf{private capital growth}} + (1-\mu) \underbrace{\Delta \log L_M}_{\textbf{labor growth}} \right)$$

$$+ \underbrace{s_T}_{\textbf{share of traditional}} \Delta \log Y_{traditional}$$

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$$\underbrace{\Delta\%\ \text{GDP}}_{3.7\%} = \underbrace{\Delta\%\ \text{Public Capital}}_{1.1\%} + \underbrace{\Delta\%\ \text{Private Capital}}_{1.7\%} + \underbrace{\Delta\%\ \text{Labor}}_{1.3\%} + \underbrace{\Delta\%\ \text{Traditional}}_{-0.4\%}$$

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$$\underbrace{\Delta\% \text{ GDP}}_{3.7\%} = \underbrace{\Delta\% \text{ Public Capital}}_{1.1\%} + \underbrace{\Delta\% \text{ Private Capital}}_{1.7\%} + \underbrace{\Delta\% \text{ Labor}}_{1.3\%} + \underbrace{\Delta\% \text{ Traditional}}_{-0.4\%}$$

$\approx 80\%$ of gains from private firms' decisions.

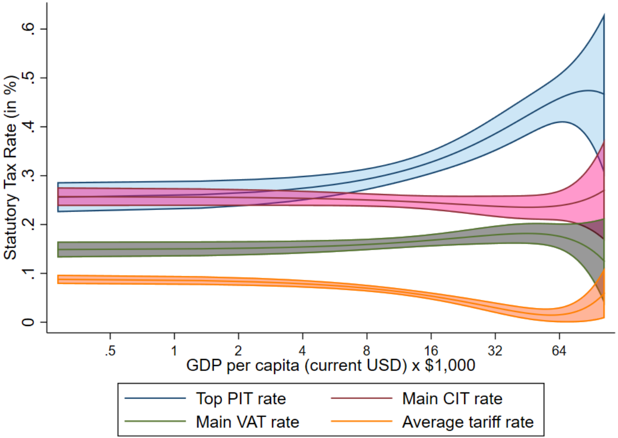
Conclusion and Future Work

- New data and measures of tax collection capacity across countries
- Richest countries 30 more collection capacity than poorest
- Quantitative Ramsey model to study tax productivity and development
- **Long-run effects of tax productivity could be substantial**
  - much larger than those suggested by simple Hulten calculation
- **Large share of gains coming from firm's decisions**
- Future work: fine-tuned calibrations to several countries, inference of tax productivity differences across countries, transition dynamics, alternative elasticities of substitution between public and private capital, etc.

Thanks!

Comments, criticism, suggestions: `abdou.ndiaye@nyu.edu`

Statutory Tax Rates in the Cross-Section of Countries



Notes: This figure plots the statutory tax rates against GDP per capita in the sample of 100 countries. The local polynomial for each type of tax rate is plotted, together with the 95 percent confidence interval.

Back to Main

Steady State

A steady state is determined by a system of 9 equations in 9 unknowns $(w_M, w_T, r, L_M, L_T, L_C, k, k^G, c)$:

$$\begin{aligned}
\text{(i) Optimal collection:} \quad & \frac{1-\beta(1-\delta_G)}{G_{L_C}(A^C, L_C)\, L_M} = \beta\Big[F_G + (1-\delta_G)\Big] - 1 + \beta\left(\frac{1-\eta}{\eta}\right) F_{L_M G} \\
\text{(ii) Private capital:} \quad & 1 = \beta\Big(F_k + 1 - \delta\Big) + \beta\frac{1-\eta}{\eta} F_{L_M,k} L_C, \\
\text{(iii) Public capital:} \quad & k_G = \frac{1}{\delta^G} G(A^C, L_C) w_M L_M, \\
\text{(iv) Return on private capital:} \quad & r = F_{M,k}, \\
\text{(v) Modern sector wage:} \quad & w_M = F_{M,L}, \\
\text{(vi) Traditional sector wage:} \quad & w_T = F_{T,L}, \\
\text{(vii) Modern labor supply:} \quad & u_{L_M} = u_c\, w_M \big(1 - G(A^C, L_C)\big), \\
\text{(viii) Traditional labor supply:} \quad & u_{L_T} = u_c\, w_T, \\
\text{(ix) Aggregate resource constraint:} \quad & c + k + w_M L_C + k_G = (1 + r - \delta)k + w_M L_M + A^T L_T.
\end{aligned}$$

Back to Main

Baseline Steady State

Variable	Parameter	Value
Modern labour	L_M	0.626
Traditional labour	L_T	0.782
Tax collectors	L_C	0.00246
Private capital	k	2.334
Public capital	k_G	1.354
GDP (output)	Y	1.501
Tax revenue	T	0.081
Average tax rate (%)	$\overline{\tau}$	16.88
Tax/GDP (%)	T/Y	5.41
Modern-sector share (%)	Y_M/Y	47.88

Back to Main

Transition Dynamics

Lemma

Assume that household preferences are separable in consumption and labor, with CRRA utility in consumption, and that the tax production function is linear in personnel $G(A^C, L_C) = A^C L_C$. Then an interior steady state exists and the optimal personnel choice over the transition (for $t > 0$) satisfies:

$$\frac{(F_G(t+1) - \delta_G) - (F_k(t+1) - \delta)}{F_k(t+1) + 1 - \delta} = \frac{1}{A^C}\left(\frac{1}{L_{M,t}} - \frac{\beta(1-\delta_G)}{L_{M,t+1}}\right).$$

Higher tax productivity A^C leads to higher levels of public capital (lower F_G). Back to Main